

Computer Vision Practical Report: Image Processing and Grayscale Conversion

OpenCV and Matplotlib Integration

1. AIM

To read an input color image using OpenCV, convert it into a grayscale image, display both the original color image and grayscale image side-by-side on the same page using Matplotlib, and save the converted grayscale image to disk.

2. PROCEDURE

1. Load the input color image ('E1-IMG.png') using OpenCV's `cv2.imread()` function.
2. Convert the image color space from BGR (OpenCV default) to RGB using `cv2.cvtColor()` so that Matplotlib displays true colors.
3. Convert the image from BGR to single-channel Grayscale format using `cv2.cvtColor()` with `cv2.COLOR_BGR2GRAY`.
4. Create a Matplotlib figure window with dimensions 10x5 inches using `plt.figure(figsize=(10, 5))`.
5. Define the first subplot (1, 2, 1) to plot the original RGB color image and set its title to 'Original Image'.
6. Define the second subplot (1, 2, 2) to plot the converted grayscale image with `colormap cmap='gray'` and set its title to 'Grayscale Image'.
7. Disable coordinate axis ticks on both subplots using `plt.axis('off')` for clean visual presentation.
8. Display both subplots side-by-side on the same figure page using `plt.show()`.
9. Save the processed grayscale image to disk as 'grayscale_image.png' using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2
import matplotlib.pyplot as plt

# Load image from file
image = cv2.imread('E1-IMG.png')

# Convert color space: BGR to RGB (for Matplotlib) and BGR to Grayscale
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Initialize plot figure for side-by-side visualization
plt.figure(figsize=(10, 5))
```

```
# Subplot 1: Original Color Image
plt.subplot(1, 2, 1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

# Subplot 2: Grayscale Image
plt.subplot(1, 2, 2)
plt.imshow(gray_image, cmap='gray')
plt.title("Grayscale Image")
plt.axis("off")

# Display plots side-by-side on the same window
plt.show()

# Save the output grayscale image
cv2.imwrite("grayscale_image.png", gray_image)
```

4. OUTPUT

The side-by-side visual output displaying the original color image alongside the converted grayscale image on the same page:



5. RESULT

The Python script executed successfully. The input image 'E1-IMG.png' was loaded, converted into a grayscale image, displayed side-by-side on the same figure with the original image, and saved to disk as 'grayscale_image.png'.